Answer Sheet

EE3014/3042 Midterm Exam **Date: 23 November, 2023**

Student Name and Number Signature

The notes and books will be kept closed. The calculator can be used. The required formulas are given.

1. (25p) A 30-kVA 8000/230-V distribution transformer has an impedance referred to the primary of 22+j98 Ω. The components of the excitation branch referred to the primary side are R_C = 100 $k\Omega$ and X_M = 20 kΩ

(a) If the primary voltage is 7970 V and the load impedance of Z_L = 1.85+ j0.75 Ω, what is the secondary voltage of the transformer? What is the voltage regulation of the transformer?

(b) If the load is disconnected and a capacitor of –j2.87 is connected in its place, what are the real values of primary and secondary voltages of the transformer? What is its voltage regulation under these conditions?

2. (25p) A simple rotating loop between curved pole faces is connected to a battery and a resistor through a switch. The resistor (R) represents the total resistance of the battery and the wire in the machine. The specifications of this machine are $R = 0.3\ \Omega$, $V_B = 120\ V$, $B = 0.25\ T$ with loop dimensions of , $l = 1.0\ m$ and $r = 0.5\ m$.

(a) Suppose a load is attached to the loop, and the resulting load torque is 10 N-m. What would the new steady-state speed be? How much power is supplied to the shaft of the machine? How much power is being supplied by the battery? Is this machine a motor or a generator?

(b) Suppose the machine is running unloaded. What would the final steady-state speed of the rotor be if the flux density were reduced to 0.20 T?

3. (25p) A 100-hp, 250-V, 1200 r/min shunt de motor with an armature resistance of 0.03 Ω and a field resistance of 41.67 Ω. The motor has compensating windings, so armature reaction can be ignored. Mechanical and core losses may be assumed to be negligible for the purposes of this problem. The motor is assumed to be driving a load with a line current of 126 A and an initial speed of 1103 r/min. To simplify the problem, assume that the amount of armature current drawn by the motor remains constant.

(a) If the machine's magnetization curve is shown in Figure 1, what is the motor's speed if the field resistance is raised to 50 Ω?

(b) How does the speed of this motor changes as a function of the field resistance R_F, assuming a constant-current load. Draw its rough graph, i.e., the plot of speed n_m versus field resistance R_F without calculation.

4. (25p) A separately excited de generator is rated at 172 kW. 430 V, 400 A, and 1800 r/min. Its magnetization curve is shown in Figure 2. This machine has the following characteristics:

$R_A = 0.05\ \Omega$ $V_F = 430$ V

$R_F = 20\ \Omega$ $N_F = 1000$ turns per pole

$R_{adj} = 0$ to $300\ \Omega$

(a) If the variable resistor R_{adj} in this generator's field circuit is adjusted to 63 Ω and the generator's prime mover is driving it at 1600 r/min, what is this generator's no-load terminal voltage?

(b) What would its voltage be if a 360-A load were connected to its terminals? Assume that the generator has compensating windings.

(c) What would its voltage be if a 360-A load were connected to its terminals, but the generator does not have compensating windings? Assume that its armature reaction at this load is 450 A • turns.

(d) How much field current would be needed to restore the terminal voltage to its no-load value? (Assume that the machine has compensating windings.) What is the required value for the resistor R_{adj} to accomplish this?

Table 1. Formulas

$P = E_A I_A$	$Q = V_s I_s sin\theta$ per phase	$P = \sqrt{3} V_s I_s cos\theta$	$F_{net} = ma$
$F = i(lxB)$	$VR = \frac{V_{P,pu} - V_{S,fl,pu}}{V_{S,fl,pu}}$	e_{ind} =(vXB).l	$\eta = \frac{P_{out}}{P_{in}} x100$
$E_A = K\emptyset w$	$Pout = w\,\tau$	$S_{base} = V_{base} I_{base}$	$Y_{EX} = G_C - jB_M$

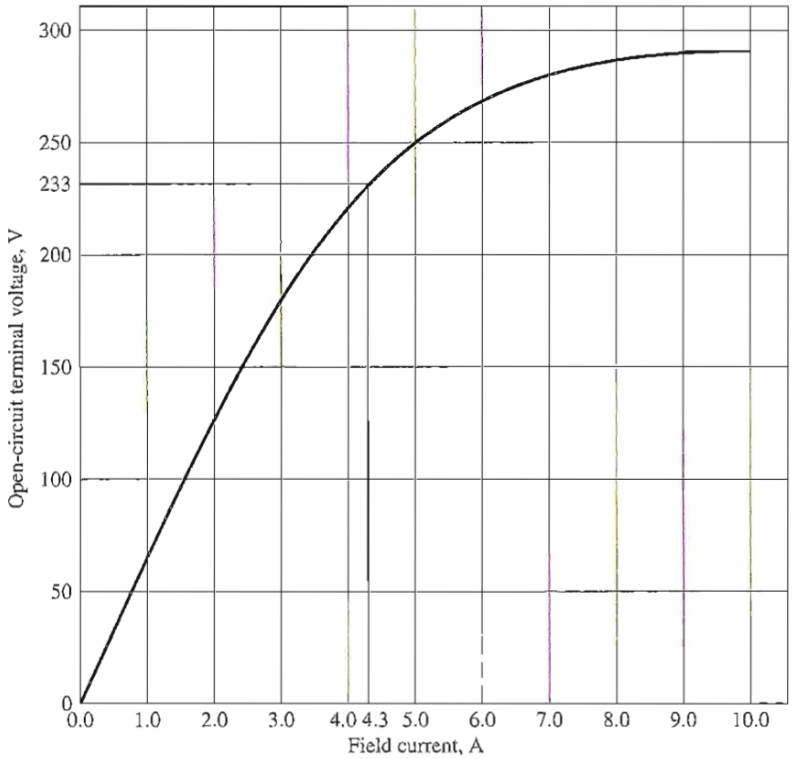


Figure 1. The magnetization curve of a typical 250-V dc motor, taken at a speed of 1200 r/min (Q3).

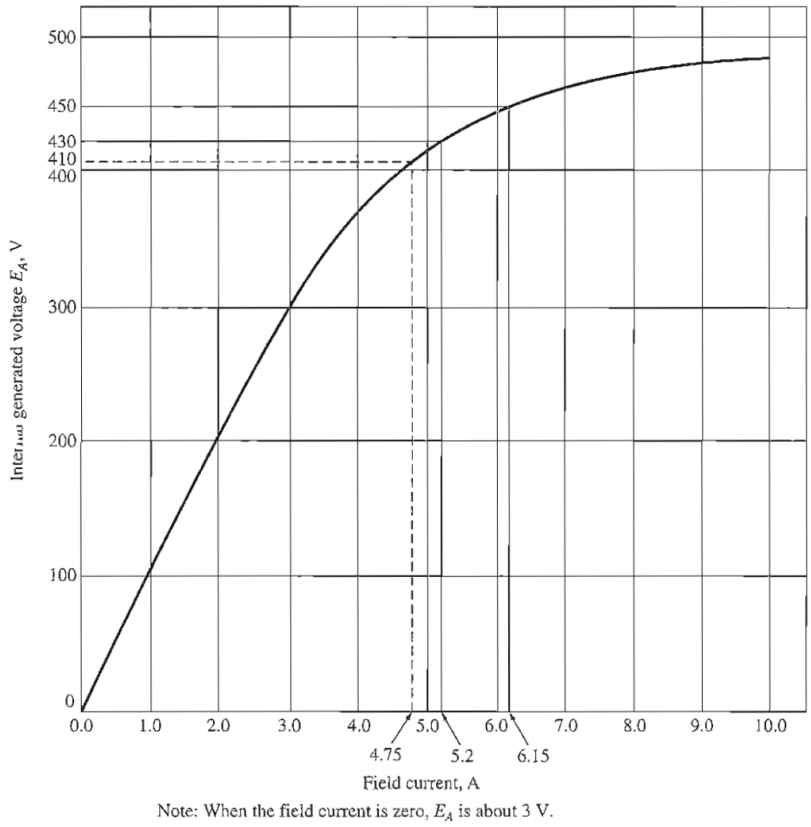


Figure 2. Magnetization curve for the generator (Q4).

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1.(25p) 30kVA 8000/230-V distribution TR

$Z_{RP} = 22 + j98\ \Omega$ (referred to primary)

$R_c = 100k\Omega$, $X_m = 20k\Omega$ } excitation branch.

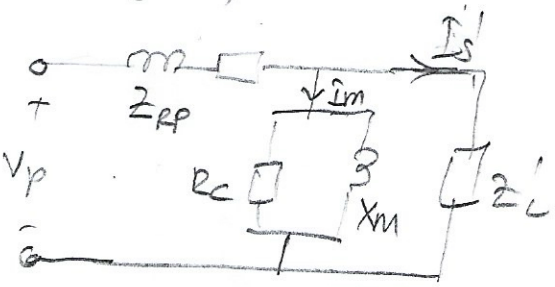
a) $V_p = 7970V$, $Z_L = 1,85 + j0,75\Omega \Rightarrow V_s = ?$, $VR = ?$

b) If $Z_L = -j\,2.87\Omega$ V_p and V_s ? $VR = ?$

SOLUTION: $V_s' = Z_L' I_s'$ → refered to the primary.

$$Z_L' = \left(\frac{8000}{230}\right)^2 Z_L = (34.783)^2 \underbrace{(1,85 + j0,75)}_{2\angle 22.1^\circ} = 2420\angle 22.1^\circ \quad 3/$$

(a)



Im is ignored.

← I_s' refered to secondary.

$$I_s' = \frac{V_p\angle 0^\circ}{Z_{RP} + Z_L'} = \frac{7970\angle 0^\circ}{\underbrace{22 + j\,98\ \Omega}_{100.44\angle 77.35^\circ} + 2420\angle 22.1^\circ} = \frac{7970\angle 0^\circ}{2478.6\angle 24} = 3,216\angle -24^\circ\ A \quad 3/$$

$$V_s' = Z_L' I_s' = (2420\angle 22.1^\circ)(3,216\angle -24^\circ) = 7783\angle -1.9^\circ\ V \quad 2/$$

$$V_s = \frac{V_s'}{a} = \frac{7783\angle -1.9^\circ}{34,783} = 223,76\angle -1.9^\circ\ V \quad 2/$$

$$VR = \frac{V_p - V_s'}{V_s'} \times 100 = \frac{7970 - 7783}{7783} \times 100 = 2.4\% \quad 3/$$

b) If $Z_L = -j\,2.87 \Rightarrow Z_L' = (34,783)^2(-j\,2.87) = -3472.3j\ \Omega$ 2/

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b) continued $I_s' = \frac{7970\angle 0°}{100.44\angle 77.35° + 3472.3\angle -90°} = \frac{7970\angle 0°}{3374.4\angle -89.63°}$

$\Rightarrow I_s' = 2.362\angle +89.63°\ A \Rightarrow V_s' = Z_L' I_s' = (3472.3\angle -90°)(2.362\angle 89.63°)$ 3/

$V_s' = 8201.6\angle -0.37$ 2/

$V_s = \frac{V_s'}{a} = \frac{8201.6\angle -0.37°}{34.783} = 235.8\angle -0.37°\ V$ 2/

$VR = \frac{V_P - V_s'}{V_s'} \times 100 = \frac{7970 - 8201.6}{8201.6} \times 100 = -2.82\ \%$ 3/

2. (25p) a) If a load torque of 10N.m is applied to the shaft of the machine, it will begin to slow down. But as w decreases, e_{ind} also decreases that causes to increase to the currents. So the τ_{ind} increases, too until $\tau_{ind} = \tau_{load}$ at a lower speed.

At steady state $\tau_{load} = \tau_{ind} = \frac{2}{\pi}\phi i$ 3/

$i = \frac{\tau_{ind}}{\frac{2}{\pi}\phi} = \frac{\tau_{ind}}{\frac{2}{\pi}BA_l} = \frac{\tau_{ind}}{\frac{2}{\pi}B(2\pi rl)/2}$

$i = \frac{10}{(0.25)(2)(0.5)(1.0)} = 40\ A$ 3/

The power transfessed to shaft $P = \tau_{ind}\, w$

The power supplied by the battery $P = V_B i$

$e_{ind} = V_B - iR = 120 - (40)(0.3) = 108\ V$ 2/

2/ $e_{ind} = \frac{2}{\pi}\phi w \Rightarrow w = \frac{\pi e_{ind}}{2 \cdot BA} = \frac{\pi e_{ind}}{2B(2\pi rl)/2} = \frac{108}{(0.25)(2)(0.5)(1.0)}$

3/ $w = 432\ rad/s$ (the speed reduces)

$P_{shaft} = \tau_{ind} w = (10)(432) = 4320\ W$ 2/

The m/c acts a motor. 1/ $\Leftarrow P_{battery\ supply} = V_B i = (120)(40) = 4800\ W$ 2/

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2) continued b) Unloaded condition, $B = 0,2\ T$

At steady state $e_{ind} = V_B$ So $i = 0$ 3/

$$e_{ind} = V_B = \frac{2}{\pi}\phi\omega = \frac{2}{\pi}BA\omega = \frac{2}{\pi}B(2\pi r l)/2\,\omega$$

2/ $$e_{ind} = V_B = 2rlB\omega \Rightarrow \omega = \frac{V_B}{2rlB} = \frac{120}{(2)(0,5)(1)(0,2)}$$

$\omega = \underline{600\ rad/s}$. 2/

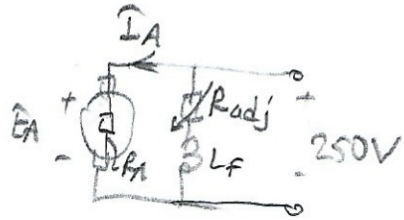
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3. (25p) 100hp, 250V, $n_{mo} = 1200$ r/min, SHUNT DC motor

$R_A = 0,03\Omega$, $R_F = 41,67\ \Omega$, $I_{AR} = 0$, $I_A = 126\ A$

with $n_{m\,initial} = 1103$ r/min. Assume I_A to be constant.

If R_F is raised to 50Ω, $n_m = ?$



$$I_{A1} = I_{L1} - I_{F1} = 126 - \frac{250}{41,67} = 120A$$ 3/

$$E_A = K'\phi n_m \Rightarrow \frac{E_{A2}}{E_{A1}} = \frac{K'\phi_2 n_{m2}}{K'\phi_1 n_{m1}} = 1$$ since I_A is constant. 4/

$$\Rightarrow n_{m2} = \frac{\phi_1}{\phi_2} n_{m1}$$ 1/

$E_{A1} = E_{A2} = V_T - I_{A1}R_A$, because I_{A1} is constant.

$E_{A1} = E_{A2} = E_A = 250 - (120)(0,03) = 246.4V$ 3/

For $R_{f1} = 41.67\Omega \Rightarrow I_{F1} = \frac{V_T}{R_{F1}} = \frac{250}{41.67} \cong 6\ A$, $n_{m1} = 1103$ r/min 2/

For $R_{F2} = 50\Omega \Rightarrow I_{F2} = \frac{V_T}{R_{F2}} = \frac{250}{50} = 5A$, $n_{m2} = ?$ 2/

Since E_A is constant $\phi = KI_F$ where ϕ changes with I_F

$$n_{m2} = \frac{KI_{F1}}{KI_{F2}} n_{m1} = \frac{6}{5}(1103) = (1.2)(1103) = 1323.6\ r/min$$ 5/7

3-continued) But, the speed changes according to the magnetization curve. In this case, we should take the ratio of E_{A1}/E_{A2} into account, instead of the ratio of I_{F1}/I_{F2}. Hence, the correct solution is

at $I_{f1} = 6A \Rightarrow E_A(I_{F1}) = 268V$, 3/

at $I_{f2} = 5A \Rightarrow E_A(I_{F2}) = 250V$ 3/

} proportional to the flux.

$$\frac{\phi_1}{\phi_2} = \frac{268V}{250V} = 1.072 \Rightarrow n_{m2} = (1,072)(1103) = 1182.4\ r/min.$$

2/

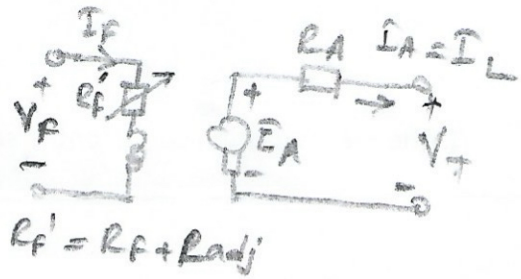
b) Part b) On page 7. 2/

4. (25p) A separately excited DC gen. rated at 172kW, 430V 400A and 1200 r/min.

$R_A = 0.05\Omega$ $V_F = 430V$

$R_F = 20\Omega$ $N_F = 1000$ turns/pole

$R_{adj} = 0$ to 300Ω



a) $R_{adj} = 63\Omega$ $n_{m1} = 1600$ r/min (prime mover driving speed)

$V_{T,no\ load} = ?$

b) V_T if $I_L = 360A$ with compensating windings

c) V_T if $I_L = 360A$ without compensating windings, $F_R = 450$ A-turns

d) $I_F = ?$ to restore V_T to $V_{T,no\ load}$, with compensating windings

The required R_{adj} ?

SOLUTION a) The total field resistance $= R_F + R_{adj} = 83\Omega$. 1/

Hence $I_F = \frac{V_F}{R_F + R_{adj}} = \frac{430}{83} = 5,18\ A$ 1/

∴ From the magnetization curve (MC) $E_{A_0} = 430V$ at 1800 r/min 2/

At 1600 r/min $\Rightarrow$ $\frac{E_A}{E_{A0}} = \frac{K\phi n_m}{K\phi n_{m_0}} \Rightarrow E_A = \frac{1600}{1800}(430) = 382.2\ V$ 2/

4. continued) $V_{T,noload} = E_A = 382.2V$ at 1600 r/min.
(no load condition) 2/

b) $I_L = 360A$ (the load connected)

$$V_T = E_A - R_A I_A = 382.2 - (0,05)(360) = 364.2V \quad 2/$$

c) $$I_F^* = I_F - \frac{I_{AR}}{N_F} = 5.18 - \frac{450}{1000} = 4,73\,A \quad 2/$$

from the MC $E_{A_0} \cong 410V$ at 1800 r/min. 1/

$$E_A = \frac{1600}{1800}(410V) = 364.4 \quad 2/ \text{ at } 1600 \text{ r/min.}$$

$$V_T = E_A - R_A I_A = 364.4 - (0,05)(360) = 346.4V \quad 2/$$
with the armature reaction

d) $$E_A = V_{T,noload} + R_A I_A = 382.2 + (0,05)(360) = 400.2V$$

The corresponding E_A at 1800 r/min 2/ at 1600 r/min

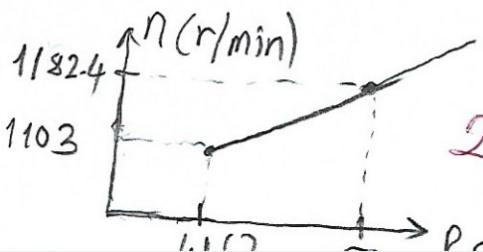
$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow E_{A0} = \left(\frac{n_0}{n_m}\right) E_A = \left(\frac{1800}{1600}\right)(400.2) = 450.23V$$

From the MC $I_F \cong 6.15\,A$ 2/ , $E_{A0} = 450.23V$ 2/

$$I_F = \frac{V_F}{R_F + R_{adj}} \Rightarrow R_F + R_{adj} = \frac{V_F}{I_F} \Rightarrow R_{adj} = \frac{V_F}{I_F} - R_F$$

$$R_{adj} = \frac{430}{6.15} - 20 \cong 50\Omega. \quad 2/$$

↓ the required set

3.b)


2/

7/7